Education and Information Technologies (2023) 28:7437–7460
https://doi.org/10.1007/s10639-022-11460-3

A systematic review of learning path recommender systems

Nur W. Rahayu[1,2] · **Ridi Ferdiana**[2] · **Sri S. Kusumawardani**[2]

Received: 3 June 2022 / Accepted: 8 November 2022 / Published online: 2 December 2022

Abstract

Learning path recommender systems are emerging. Given the popularity of ontology/knowledge-based systems in adaptive learning, this work reviews learning path in ontology-based recommender systems. The review covers recommendation trends, ontology use, recommendation process, recommendation technique, contributing factors, and recommender evaluations. A total of 12,972 articles published between 2010 and 2020 were identified in the initial search across five major databases, and 9 of them are considered in this work. Currently, student model, learning objects, learning activities, and external environment are contributing factors for recommending learning object sequence. We also found that the current trend for LP recommendations process is semi-dynamic and dynamic. Semi-dynamic learning path are started by a pre-set path, while dynamic learning path is flexible from the first step and intended for personal use. The recommendation process itself has four phases: predelivery of the first learning object, current learning object delivery, learning object postdelivery, and predelivery of the next learning object. The current recommendation technique collaborates ontology and several techniques, such as Bayesian networks, data mining, and other artificial intelligence technique. To evaluate performance, learning path recommender systems use real students, control groups in parallel or sequential experiments, and student satisfaction surveys. Ontology could work with knowledge representation instruments, educational psychology, and evolutionary computation to create a future dynamic learning path in adaptive learning environment.

Keywords Adaptive learning · Dynamic learning path · Learning object · Learning object sequence · Ontology-based recommender systems

Abbreviations

ACO	Ant Colony Optimization
ADL	Advanced Distributed Learning

✉ Sri S. Kusumawardani
suning@ugm.ac.id

Extended author information available on the last page of the article

AR	Augmented Reality
CAI	Computer-Aided Instruction
CDT	Context Dimension Tree
GA	Genetic Algorithm
ITS	Intelligent Tutoring System
LMS	Learning Management System
LP	Learning Path
LO	Learning Object
PRISMA	Preferred Reporting Items for Systematic Reviewsand Meta-Analyses
VLE	Virtual Learning Environment

1 Introduction

E-learning recommender systems have become not only a necessity but also a trend in the current information overload era (Moreno-Marcos et al., 2019). Recommendations are used to maintain student engagement, help students with problems (R. Huang et al., 2019), and improve academic achievement (Harley et al., 2018). However, recommendations are more difficult to create in the e-learning domain than in the e-commerce or e-government domain because students' interests, expectations, and abilities vary and change over time (Fidalgo-Blanco et al., 2015; George & Lal, 2019; R. Huang et al., 2019).

Mostly, e-learning recommender system personalize learning objects (LO), but learning path, peer, and feedback are emerging recommendation items (Manouselis et al., 2013; Rahayu et al., 2022). As a growing topic, learning path (LP) refer to either course sequences in a particular study program (Iatrellis et al., 2019) or learning object sequences within a course (Machado et al., 2021; Premlatha & Geetha, 2015). The latter definition was adopted in the current study, in which optimal LPs were found to be helpful reducing students' cognitive surplus and increasing their learning efficiency.

Learning object sequences in e-learning are clearly different from those in classical and offline classes because in the e-learning, adaptive sequences are constructed based on each student's learning needs, learning history, and curriculum information. Therefore, secondary studies show that LP recommender systems usually deploy machine learning and soft computing techniques, e.g., collaborative filtering, fuzzy logic, association rule mining, heuristic algorithms, evolutionary computing, or hybrid (Kardan et al., 2015; Machado et al., 2021; Premlatha & Geetha, 2015). However, web technology has made ontology/knowledge-based recommender systems more prevalent (George & Lal, 2019; Monti et al., 2020; Tarus et al., 2018). Therefore, LP positioning in ontology-based recommender systems needs to be detailed.

This review addresses the following research questions:

RQ1: To what extent have ontologies in learning path recommender systems been used?

RQ2: What are the contributing factors in the learning path recommendation process?
RQ3: What are the state-of-the-art techniques for recommending optimal learning paths?
RQ4: How are learning path recommender systems evaluated?

This study aims to find and classify primary studies related to learning path in ontology-based recommender systems. The literature search involves broad automated searches of journal articles published in Springer Link, IEEE Xplore, Scopus, Science Direct, and ERIC databases from 2010 to 2020. The current review covers recommendation trends, ontology use, recommendation process, recommendation technique, contributing factors, and recommender evaluations. By reviewing the latest LP recommender systems, this study can enrich the knowledge base for the research community specializing in ontology and semantic web. Instructors and system developers can also take advantage of the findings to enhance their knowledge and increase students' learning efficiency.

The remaining sections of this paper continue as follows: Section 2 provides a theoretical background. Sections 3 and 4 respectively highlight a review of related secondary studies and the research methodology. In Section 5, we present the extraction results and analysis to answer RQ1-RQ4. Section 6 is dedicated to discussing implications, research agendas and limitations. Finally, Section 7 concludes the current study.

2 Theoretical background

2.1 Recommender system

Recommender systems choose the best and most relevant items for users. Historically, recommender systems emerged from collaborative filtering systems (Resnick & Varian, 1997). The definition is also expanding, but the recommender system focuses on two key characteristics: (1) individualized and (2) "interesting and useful" (Burke, 2002).

Expert classifications vary as well. Recommender systems can be classified according to recommendation technique including content-based, collaborative, and hybrid approaches (Adomavicius & Tuzhilin, 2005). In addition to these three categories, there are also knowledge-based, fuzzy set-based, social network-based, trust-based, context-based awareness, and group recommendations approach (Lu et al., 2015). Recommender systems can also be put into two broad groups based on interaction/attribute data and context domains. Collaborative filtering and content- and knowledge-based recommendation are examples of the first type, whereas location, time, and social data are examples of the second category (Aggarwal, 2016).

Recommender systems can also be classified by application domains, such as e-government, e-commerce, and e-learning. Therefore, such systems are used to recommend products, news, friends, and advertisements for relevant users. In the e-learning domain, recommendations are used to maintain student engagement, improve academic achievement, and help overcome student's problems. However, studies have shown that recommendations in the e-learning domain are difficult to provide because students' interests, expectations, and abilities vary and change over time (Fidalgo-Blanco et al., 2015; George & Lal, 2019; R. Huang et al., 2019).

2.2 Ontology-based recommender system

Recommender systems in e-learning are expanding. Finding learning objects, finding peers, finding learning paths, predicting student performance, and recommending courses are all examples of supported tasks for e-learning recommender systems (Manouselis et al., 2013). Recommender systems can also be described by how they handle student models, domain models, and personalization, such as by using collaborative filtering, classifier-based models, machine learning, and knowledge-based models that use ontology.

An ontology itself is defined as a formal, explicit specification of a shared conceptualization (Studer et al., 1998). Previous study discovered that ontology/knowledge-based systems are the dominant techniques in e-learning recommender systems because of their capabilities in cold-start problem solving, sparsity rating, and overspecialized recommendations (George & Lal, 2019; Tarus et al., 2018). The cold-start condition refers to the difficulty in providing recommendations to new student users whose preferences have yet to be known. Meanwhile, the sparsity rating indicates a common situation in e-learning, i.e., very few students give ratings to LOs. These two problems create additional drawback, i.e., low accuracy and limited recommendation items (*overspecialized recommendation*). A success story was shown in a previous study: a recommender system that used ontology to profile users increased the accuracy of its recommendations by 7–15% by using only is-a relationships. (Middleton et al., 2009).

Ontology is widely used in various subjects, created by expert ontologists and instructors (Boyce & Pahl, 2007), and developed using methodologies such as NeOn methodology (Mariño et al., 2018), (Yago et al., 2018) and On-To-Knowledge (Labib et al., 2017). There are three types of ontologies in learning technology, i.e., ontologies for student models, LOs, and learning activities (Devedzic, 2006). Recently, ontology-based recommender systems are now hybrid, collaborating ontology with other methodologies and even multidisciplinary (Rahayu et al., 2022).

2.3 Learning path

Learning path represent a route taken by a student in studying the available LOs (Premlatha & Geetha, 2015). Synonyms for LP include content or learning content sequencing (Zapata-Ros, 2006), sequence generation (Rasheed & Wahid, 2019), curriculum sequence (Iglesias et al., 2004; Machado et al., 2021), study path, and instructional sequence (Pepin & Kock, 2021). A course's learning path is usually established by a teacher or researcher in the relevant field and is the same for all students. This condition then changes to an adaptable learning path due to technological advances and student variances in knowledge, preferences, and learning speed. According to education literature, a learning path for an entire class is known as the micro-level curriculum, while the learning path for an individual student is known as the nano-level curriculum (Leyendecker, 2012). This study employs the term learning path to describe the adaptive sequencing of learning objects for each individual student (nano-level curriculum).

As a learning route, optimal LPs help students to study efficiently. Therefore, the absence of adaptive LPs increases the learning achievement gap among students in a class (Huang et al., 2007) and decreases learning satisfaction (Katuk et al., 2013), learning effectiveness (Hsieh & Wang, 2010; Huang et al., 2007), and learning time efficiency.

3 Related works

There are few review articles that address the learning path or curriculum sequencing from a computer science perspective. Learning paths of open educational resources (OER) are discussed in a study (Siren & Tzerpos, 2022). This research examines path creation using standard linear models or various predetermined paths to meet learning goals. The second type demands for a recommender algorithm based on prerequisite concepts, similarity measurements, and shortest path. However, the study did not address algorithms, student model or other variables that could influence the LP recommendation process.

Variations in learning path recommendation technique are discussed in another study (Nabizadeh et al., 2020). The authors differentiated between course generation (recommending a full path to a student in a single recommendation) and course sequencing (recommending a path to a student in stages based on the user's progress). By the end of 2020, most studies were focused with course generation, not course sequences, according to the study. Therefore, course sequencing is currently a growing issue. The article also briefly discusses several algorithms for generating course sequences, including Association Link Network, Item Response Theory, evolutionary computation, and Bayes theorem. However, this study did not synthesize the pattern of recommendation process.

A study that compares metaheuristic algorithms also talks in depth about algorithmic solutions (Machado et al., 2021). Because LO sequencing was categorized as an NP-hard problem (nondeterministic polynomial time), these algorithms were selected. This paper addresses Ant Colony Optimization (ACO), Genetic Algorithm (GA), PSO, and others. This study has also considered several parameters for student and domain models but has not addressed additional relevant factors or ontology use. Therefore, our research attempts to map ontology use in LP recommender systems, focusing on trends, contributing factors, and the recommendations process.

4 Methodology

The current research used the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram (Page et al., 2021). This study uses the latest version of PRISMA ("the PRISMA 2020 statement") as a reference for the criteria. This section describes the details of the literature search, article selection, and data extraction processes.

4.1 Literature search

Article searches were carried out using a broad automated search on five digital databases, i.e., Springer Link, IEEE Xplore, Scopus, Science Direct, and ERIC. A search was performed on all metadata (if available) with the search keywords: *(adapt* OR personali* OR recommend* OR intelligent*) AND (student OR learner) AND* (*path* OR sequence OR order*) AND (ontology)*." A total of 12,972 articles were obtained: 10,133 articles from Springer Link, 2,725 from IEEE Xplore, 94 from Scopus, 18 from Science Direct, and 2 from ERIC.

4.2 Article selection

For inclusion in the study, the reviewed papers were subjected to a selection procedure, which consisted of inclusion and exclusion phases. The initial selection was used to sort out duplicate articles and to select articles that are in the form of journal articles and written in English. A total of 341 articles obtained from this initial selection were then processed in the inclusion stage.

The inclusion stage required the articles to be full papers about the e-learning domain that are in the form of primary studies on LPs and were published between 2010 and 2020. This process was performed by examining the titles and abstracts of the articles. At the end of this stage, 26 articles were obtained.

The exclusion stage filtered the articles through a full-text review to ensure that the articles addressed the LP recommendations technically. A total of 17 articles were eliminated at this stage because they are not focused on LPs, they only choose LOs for only one learning session, they discuss LPs conceptually or they explain static LP. Figure 1 shows the stages used in this study, from the search to the extraction process. Finally, 9 articles were successfully selected for the extraction stage.

4.3 Data extraction process

All articles that passed the selection phase were extracted using some criteria. In summary, four groups of data were extracted:

(1) Technology: model/system's name; research output (model, prototype); type of learning technology; system platform; ontology collaborator.
(2) Ontology model: ontology language; ontology models; student model (student profile, student model's data, learning style); LO model (type, metadata).

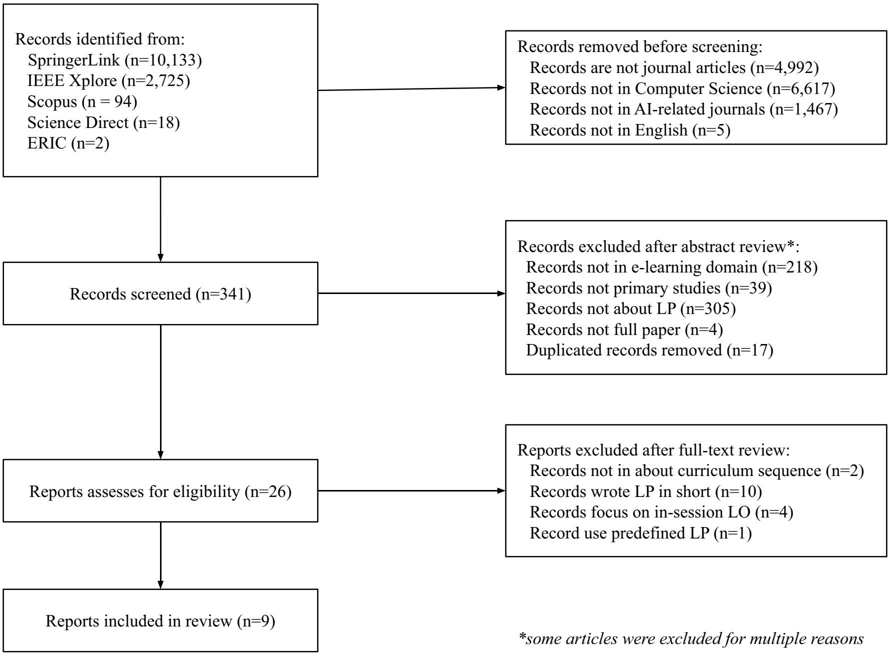

Fig. 1 PRISMA flow diagram for article selection

(3) LP method: recommendation process (static, semi-dynamic, dynamic); curriculum sequencing technique; pedagogical approach (competency-based approach, goal-based approach); factors affecting LP; recommendation technique.
(4) LP evaluation: case study (if any); country; test scale (group, institution); participant type; number of participants; course/topic/domain; course/experiment duration; control group (if any); study variables; evaluation method; baseline (if any); evaluation result.

5 Results

The learning path recommender system has been applied in various development stages, environments, and disciplines. Seven articles discussed e-learning in the form of prototypes while the remaining two were model studies. Most systems do not describe the platforms explicitly, but two studies did declare using a web platform (Capuano & Toti, 2019; Colace et al., 2020). The popular e-learning technologies in the articles were Computer-Aided Instruction (CAI) and Intelligent Tutoring System (ITS). CAI was used by students to learn course materials (Colace & Santo, 2010; Jeng & Huang, 2019) and by adult learners to study cultural heritage (Colace et al., 2020) or law (Capuano & Toti, 2019). ITS was available for students of business (Vesin et al., 2013) and computer majors (Jevremovic et al., 2017).

5.1 RQ1: To what extent have ontologies in LPs recommender systems been used?

One of the nine reviewed studies relies solely on ontology, but the remainder are hybrid, i.e., they collaborate ontology with other fields. Approaches to problem-based learning (Jevremovic et al., 2017), competency-based teaching (Hnida et al., 2014), and content presentation by storytelling (Capuano & Toti, 2019) were examples of the explicit use of the education discipline. We also noted ontology collaboration with other artificial intelligence techniques, i.e., natural language processing, context dimension tree (CDT), data clustering, collaborative filtering, rule association mining, and multiagent technique. In addition, schema theory, Petri net learning map, and Bayesian probability were also used.

The ontology has been used in reviewed studies either as a student or LO model. The results are listed below.

1. Ontology model of students (student model)

   The student ontology usually includes a profile, learning style, and performance (Grubišić et al., 2013; Kurilovas et al., 2014). Profiles could use the standard IMS Learner Information Package (Hnida et al., 2014) or custom metadata (such as age (Colace et al., 2020) and initial proficiency level (Bouhdidi et al., 2013)). Learning style was rarely mentioned explicitly in the reviewed studies; thus, only two model types were identified: the Felder–Silverman model (Bouhdidi et al., 2013) and Dunn and Dunn model (Vesin et al., 2013). Most primary studies measured students' performance level by using assessments. The assessment results were stored in the student model either as a single score for each concept (Colace & Santo, 2010) or as an aggregate score (Bouhdidi et al., 2013; Jeng & Huang, 2019; Romero et al., 2019; Vesin et al., 2013). An ontology was also used to combine assessment outcomes with other learning activities (Jevremovic et al., 2017).
2. Ontology model of learning object

   Learning objects (LOs) or learning resources are entities/resources that can be used or referenced during technology-assisted learning (R. Huang et al., 2019; Kurilovas & Juskeviciene, 2015; Labib et al., 2017). In the reviewed studies, LOs were also called schema (Jeng & Huang, 2019), node (Romero et al., 2019), lesson (Vesin et al., 2013), hypermedia unit (Bouhdidi et al., 2013), or concept (Jevremovic et al., 2017). The ontology model stored learning objects by their standard properties (such as title, description, keywords, resources, and state) as well as additional properties, like prior knowledge, suggested next LO (Jeng & Huang, 2019), additional material (Vesin et al., 2013), and educational/pedagogical level (Bouhdidi et al., 2013). LOs vary but can be grouped into two types:

   a Main content, such as files, slides, links, images, AR, game narration, tutorials, storytelling scripts, sample questions, explanatory audio, and explanatory videos.
   b Assessment, such as exam questions, game quizzes (Colace et al., 2020), practice questions (Jevremovic et al., 2017), and remedial assessment (Jevremovic et al., 2017; Romero et al., 2019).

5.2 RQ2: What are the contributing factors that affect the LP recommendation process?

The extraction results indicated that the LP recommendation process is influenced not only by the students and LOs but also by the learning activities and context (external environment). All contributing factors of LPs recommendation are shown in the Table 1.

Only one study explicitly used the Advanced Distributed Learning Sharable Content Object Reference Model (ADL SCORM) standard as LO metadata (Vesin et al., 2013); the other studies customized metadata in the following form:

a Content-related metadata: descriptions, content types, relationships between contents according to learning taxonomy (Jeng & Huang, 2019; Vesin et al., 2013), and actual learning materials.
b Metadata related to pedagogy: learning objectives, teaching methods, level of interaction, level of difficulty, and keywords (Bouhdidi et al., 2013).
c Technical-related metadata: content format, release date, and Uniform Resource Identifier (Bouhdidi et al., 2013).

5.3 RQ3: What are the state-of-the-art techniques for recommending optimal learning paths?

Adaptive LPs could be categorized based on some approaches. Curriculum sequence techniques classify LPs as individual and social. Individual sequencing bases the optimal learning path on the individual student, while social sequencing bases it on the collective path and group performance. The path based on learning objectives was named a goal-based approach, while the route given according to student achievement was called a competence-based approach. The extraction shows that most studies use individual sequencing, and that goal-based and competency-based LPs are balanced (Table 2).

The current study classified LPs by recommendation process: static, semi-dynamic and dynamic. Static LPs bring equal LPs for students. Such paths could also be carried out by keyword extraction followed by LO selection. Afterwards, the system arranged the LO sequences according to the predefined relationship among LOs. Semi-dynamic LPs may be either (1) initiated with a default path and then dynamically modified to adapt to student conditions, or (2) personalized learning paths for a group of students. The initial path may be selected by the instructor's expertise or prior learning patterns (Colace & Santo, 2010). Semi-dynamic LP recommendations may be generated just from domain ontology (Bouhdidi et al., 2013; Vesin et al., 2013); collaboration of ontology and Bayesian networks (Colace & Santo, 2010); or combination of ontology, Bayesian networks, and CDT (Colace et al., 2020). Table 3 shows a summary of the semi-dynamic LP recommendation process.

Dynamic learning paths provide students flexibility from the start. The pretest materials of dynamic LPs are varied, e.g., intellectual quotient (Vesin et al., 2013), basic problem-solving skills (Vesin et al., 2013), and prerequisite courses

Table 1 Learning path variables

No	Variable	Component
1.	Student	Profile (preference (Colace et al., 2020; Vesin et al., 2013), age group (Colace et al., 2020), discipline (Bouhdidi et al., 2013)); Learner style (Bouhdidi et al., 2013; Vesin et al., 2013); Ability (level of mastery (Bouhdidi et al., 2013; Hnida et al., 2014; Jeng & Huang, 2019; Jevremovic et al., 2017; Romero et al., 2019; Vesin et al., 2013) Learning progress (Romero et al., 2019)
2.	LO	Content (relationship among concepts (Capuano & Toti, 2019), threshold score (Jeng & Huang, 2019), keyword (Capuano & Toti, 2019)); Pedagogy (learning objective) (Romero et al., 2019) Content format (Bouhdidi et al., 2013))
3.	Learning activity	Number of LOs in the internal system; number of LOs outside the system; number of message exchanges among students; exam time; number of assessment trials (Jevremovic et al., 2017)
4.	Context or external environment	Opinions from similar students (Vesin et al., 2013); study hours (Colace et al., 2020); weather (Colace et al., 2020)

Table 2 Learning path classification

Authors	Curriculum sequencing*		Pedagogical approach**		Recommendation process		
	Individual sequencing	Social sequencing	Goal-based approach	Competency-based approach	Static	Semi dynamic	Dynamic
Bouhdidi et al. (2013)	✓	-	✓	-	-	✓	-
Capuano and Toti (2019)	✓	-	✓	-	✓	-	-
Colace and Santo (2010)	✓	-	✓	-	-	✓	-
Colace et al. (2020)	✓	-	✓	-	-	✓	-
Hnida et al. (2014)	-	✓	-	✓	-	✓	-
Jeng and Huang (2019)	-	✓	-	✓	-	-	✓
Jevremovic et al. (2017)	✓	-	-	✓	-	✓	-
Romero et al. (2019)	✓	-	-	✓	-	-	✓
Vesin et al. (2013)	-	✓	-	✓	-	-	✓

* the classification derived from (Al-Muhaideb & Menai, 2011)

** the category based on (Hnida et al., 2014)

Table 3 Process of semi-dynamic LP recommendations

Phase	Technical details		
	Instructor	System	Student
Predelivery of the first LO	Compile a learning map (Hnida et al., 2014; Jeng & Huang, 2019)	Instantiate first LOs (Romero et al., 2019); choose LP according to the pretest results (Hnida et al., 2014)	Take pretest (Hnida et al., 2014; Jevremovic et al., 2017)
LO delivery	-	Monitor student activity/performance (Jevremovic et al., 2017)	Study LO, take assessment, study new LO (Jeng & Huang, 2019)
LO postdelivery	-	Assess scores, update student models, and assess related learning activities; provide feedback and compare student achievements (Jevremovic et al., 2017); compare student performance and overall LO (Jeng & Huang, 2019);	Receive learning progress comparison (Jevremovic et al., 2017)
Predelivery of next LO	-	Check LO that has been and has not been mastered (Jeng & Huang, 2019; Jevremovic et al., 2017); check next LO in the predefined sequence (Jeng & Huang, 2019; Romero et al., 2019); create a new assessment (Jevremovic et al., 2017); move students to another cluster (Hnida et al., 2014)	Prepare for remedial (Jevremovic et al., 2017; Romero et al., 2019)

Table 4 Phases of dynamic LP recommendations

Phase	Technical details	
	System	Student
Predelivery of first LO	Cluster students (Vesin et al., 2013); check learning objectives (Bouhdidi et al., 2013); select LO according to student profile and preference (Bouhdidi et al., 2013); calculate the probability success of each LO (Colace & Santo, 2010; Bouhdidi et al., 2013); select a set of questions (Colace & Santo, 2010)	Take pretest (Colace & Santo, 2010)
LO delivery	Monitor student activity/performance (Vesin et al., 2013)	Study LO and take the assessment
LO postdelivery	Assess scores, update student model, update cluster data (Vesin et al., 2013); send student progress reports to the instructor (Colace et al., 2020)	Receive the learning progress report (Colace et al., 2020)
Predelivery of next LO	Check the next LO according to the predetermined order (Vesin et al., 2013); select next LO from similar students (Vesin et al., 2013); predict the probability success of the next LO (Colace et al., 2020; Colace & Santo, 2010; Bouhdidi et al., 2013); create a new LP, combine the ontology graph and CDT (Colace et al., 2020)	

(Colace & Santo, 2010; Jevremovic et al., 2017). Pretest results were then measured using a predetermined threshold score. Table 4 detailed the dynamic LP recommendation process.

In the phase of "Predelivery of next LO", the system is responsible for recommending the next LO by using several techniques, such as:

a Bayesian network for predicting learning success based on the probability of each LO either from previous students' performance or from questionnaires (Colace et al., 2020; Colace & Santo, 2010; Bouhdidi et al., 2013);
b Petri net that contains places and transitions to track and record the LO learning status and portfolio (Jeng & Huang, 2019);
c Association rule mining on LO and learning activities, which are selected by students and ranked based on similar students (Vesin et al., 2013);
d Collaboration filtering based on navigation patterns and ratings for each completed LO of similar students (Vesin et al., 2013);
e CDT to model the context of the ontology model in a decision tree (Colace et al., 2020).

5.4 RQ4: How are learning path recommender systems evaluated?

Six reviewed papers examined LP recommender systems by using descriptive statistical methods and questionnaires. All prototypes were tested by students/participants and experts (Jeng & Huang, 2019), and four of them were compared with control group. The control groups were used either in the parallel or serial experiments as described below.

a *Parallel experiments* (at the same period) involving a comparison with one control group who were taught by conventional methods (Vesin et al., 2013) or with three control groups (conventional method, ontology-based recommendations, and Petri net-based recommendation only) (Jeng & Huang, 2019).
b *Serial experiments* (compared with previous semester) by using similar (Colace & Santo, 2010) or different (Jevremovic et al., 2017) recommendation technique.

Most systems were implemented and tested in computer science subjects, such as Introduction to Computer Science, Java programming, computer networking, and mobile application development courses. However, the treatments were relatively different (see complete data in Table 5). The current review found that the recommended LPs were tested dynamically by real users as follows:

1. Student achievement was evaluated by comparing either the learning curves per LO (Vesin et al., 2013) or the middle and final scores (Colace & Santo, 2010; Jeng & Huang, 2019; Jevremovic et al., 2017).
2. The recommender system calculated the efficiency of each LO and the effectiveness of completion per LO (Jeng & Huang, 2019; Vesin et al., 2013).

Table 5 Details of LP evaluation

No.	Study	Recommendation technique	Number of participants	Duration per experiment	Test period	Number of control groups	Activity before study
1.	Jeng and Huang (2019)	Ontology, schema theory, Petri-Net learning map	86	4 months	Parallel	3	N/A
2.	Colace and Santo (2010)	Ontology, Bayesian network	71–100	1 semester	Serial	1	Pre-test of related subjects
3.	Vesin et al. (2013)	Ontology, student clustering, collaboration filtering, association rule mining	70	4 months	Parallel	1	Pre-test of basic academic and computer literacy skills
4.	Jevremovic et al. (2017)	Ontology, problem-based learning	84–93	1 semester	Serial	3	N/A

3. The recommender system assessed students' perceived satisfaction of speed, depth of recommendation, LP accuracy, reusability (Colace et al., 2020; Jeng & Huang, 2019; Vesin et al., 2013), and effectiveness (Capuano & Toti, 2019; Colace & Santo, 2010; Vesin et al., 2013).

All experiments passed. The first study (in Table 5) showed that the participants' perceptions were positive, and e-learning experts and teachers agreed that the recommender system can help students and teachers. According to the second study, student performance improved more in groups using the proposed tool. The third study indicated that the Experimental group had a smaller learning curve and shorter completion time than the Control group. The last study indicated that using a recommender system improved score the most.

6 Discussion

6.1 Implications

6.1.1 Design of learning path recommender systems

Learning path recommendation process affected by students, LOs, learning activities and external environment. The finding supports previous study that identified student profiles, learning style, levels of knowledge, preferences, and learning speed (Premlatha & Geetha, 2015). The classification of recommendation process show that semi-dynamic LPs are started by a pre-set path, while dynamic LP is flexible from the first step and intended for personal use. Currently, semi-dynamic and dynamic LPs are the latest trend (Porcel et al., 2018; Salehi et al., 2013).

The use of ontology in e-learning was a trend in the early 20th century (Magnisalis et al., 2011), and it became one of the dominant techniques for recommendation (Tarus et al., 2018). However, some studies described a new trend of hybrid techniques, such as ontology collaborated with various artificial intelligence techniques (George & Lal, 2019). Therefore, the current study extends the findings of previous studies because ontology collaborated with education or educational psychology.

This study also found that personalized LPs can apply to individuals or groups of similar students. The result strengthens previous classifications of curriculum sequence technique (Al-Muhaideb & Menai, 2011). We also extend previous LP solutions that use LO clustering by collaborative tagging and folksonomy (George & Lal, 2019; Klašnja-Milicevic et al., 2015). Current grouping methods used age, learning style, preference, and performance. The methods could be extended to include variables from educational psychology, such as motivational variables (e.g., motivation, attitude, and anxiety) (Dahman & Dahman, 2020) or engagement level (i.e., groups of very diligent, consistent, or sporadic students) (Moreno-Marcos et al., 2019).

The LO attributes recognized as contributing factors of LPs were learning objectives, relationships among LOs, and content format. This finding strengthens previous literature explaining learning objectives (Premlatha & Geetha,

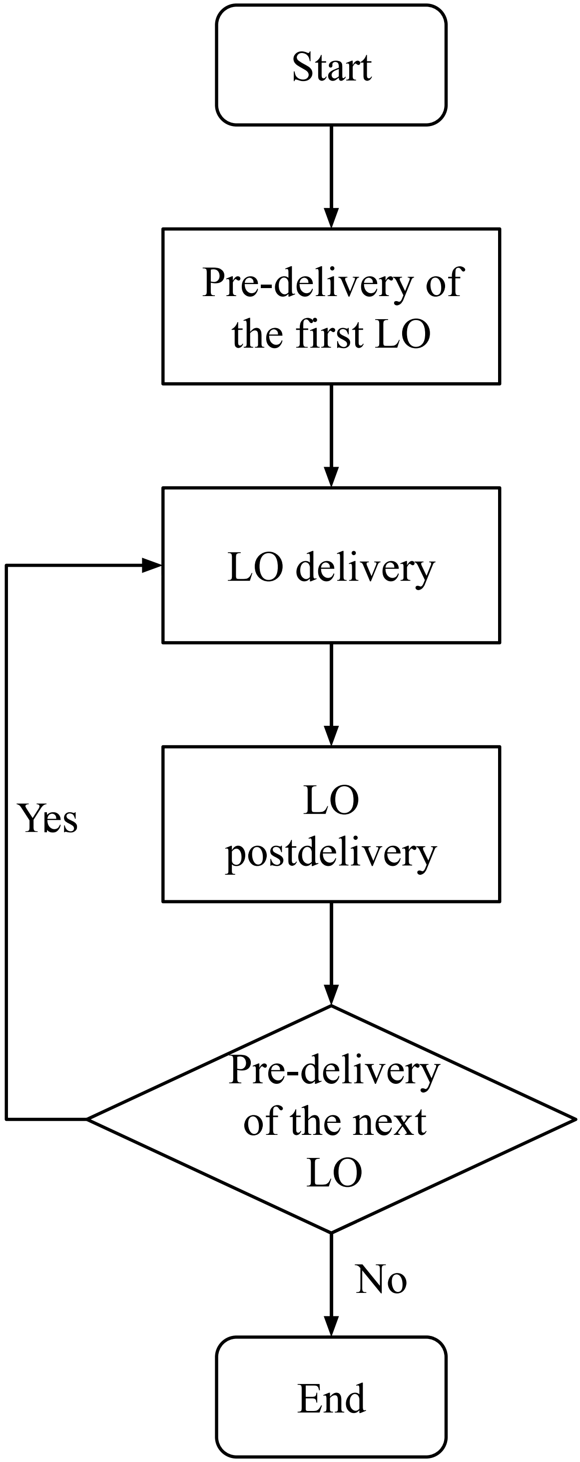

Fig. 2 Process of LP recommendation

2015), which are, in turn, supports for linked data use (Cortinovis et al., 2019). The current study also added several variables of learning activities and external conditions, such as the number of LOs accessed inside and outside the system, exam time, and study hours (Table 1).

6.1.2 Recommendation phases

The LP recommendation process starts before students learn (Tables 3 and 4). The interaction pattern has four phases, i.e., (i) predelivery of the first LO, (ii) LO delivery, (iii) LO postdelivery, and (iv) predelivery of the next LO (Fig. 2).

The system and the students (as well as instructors) present a reciprocal interaction in the first phase. In the second phase, students study the LOs, and the system monitors several aspects (e.g., LO type, number of LOs, study hours, and understanding level). The recommender system's capability can be improved by recording implicit data, such as motivation (Abdullatif & Velázquez-Iturbide, 2020), cognitive load, and engagement level (Truong, 2016).

In the third phase, the condition varies among studies. But at a minimum level, the system assesses the student performance and updates the student model. The process of updating the student model provides a greater computational burden than data addition (especially if performance conflicts exist (Clemente et al., 2014)); thus, this phase should prioritize the use of a highly efficient algorithm. In the fourth phase, the system is responsible for recommending the next LO.

To do so, there were four LP problem approaches, i.e., traveling salesman problem, constraint satisfaction problem, multicriteria or multi objective optimization, and graph (Muhammad et al., 2016). Therefore, many machine learning/soft computing techniques were used in previous studies, such as Bayesian network, Petri Net, ACO, GA, Forward Greedy algorithm, hill climbing, heuristic algorithms, memetic algorithms, PSO, and SOM (Muhammad et al., 2016; Premlatha & Geetha, 2015). The collaboration between ontology and such algorithm would probably evolve.

6.1.3 Learning path evaluation

System testing is an important phase in the information systems development methodology (Laudon & Laudon, 2012). As recommender systems are a subset of information systems, the reviewed studies also included testing. Two testing types were used, i.e., static testing (without executing the program) and dynamic testing (by executing the real input (ISO/IEC/IEEE, 2017).

The existence of assessment of students' perceived satisfaction and control groups also emerged as a trend. This finding broadens the scope of the discussion from previous secondary studies, which mostly focused on the comparison of computation techniques (Al-Yahya et al., 2015; George & Lal, 2019; Premlatha & Geetha, 2015; Tarus et al., 2018).

6.2 Research agenda

Learning path recommendations are less popular than LO recommendations (Tarus et al., 2018). However, LP recommendations will likely expand as e-learning technology advances. The reviewed studies already used LPs in CAI, ITS, and VLE. This finding is in line with previous study (Tibaná-herrera et al., 2018), which listed Learning Management System (LMS), e-assessment, VLE, Augmented Reality (AR), and ITS as the top five popular learning technologies. Therefore, LP recommendations could have great opportunities to be used in LMS and AR.

Ontology and knowledge-based systems are increasingly popular in e-learning recommender systems (Tarus et al., 2018). Learning technology, student models, learning domains, learning activities, and external factors also change dynamically. However, a semantic reasoner (i.e., a built-in inference engine for ontology-based systems) cannot be used to compile LPs because of LPs complexity. New hybrid strategies may be developed, such as combining ontology with knowledge representation instruments (concept maps or knowledge maps (Buitrago & Chiappe, 2019)), utilizing software agents (Bremgartner, 2015), or applying evolutionary computation algorithms (Premlatha & Geetha, 2015).

Table 5 show that testing was usually performed with 70–90 students in the span of four months. The existence of the pretest and the number of control groups were factors to be considered in evaluating LP recommendations. Testing types and scale could be expanded in the future, for example, by applying static testing, using learning domains other than computer science subjects, applying efficiency evaluation (e.g., learning time, completion time) and utilizing effectiveness evaluation (e.g., frequency of LOs access, assistance rate) (ISO/IEC/IEEE, 2017).

6.3 Limitations

Due to the nature of the review, our work has a few limitations. First, as students learn more on their own, adaptive LP could be nonlinear instead of linear. Nonlinear means that students can begin learning from any LO, choose their own LP, and even skip over LOs they've already mastered (Rahayu et al., 2021; Robberecht, 2007). Unfortunately, this study found no primary studies concerning nonlinear LP recommendation. Therefore, future research needs to provide a more complete picture. Second, this review is restricted to English-language literature; hence, further papers written in other languages would deepen the research findings. However, these limitations don't reduce the value of this research; in fact, our findings contribute the basis for future LPs recommender systems.

7 Conclusion

Learning paths are useful recommendation items for determining the best learning object sequences for corresponding students. This literature review aimed to investigate the latest research on learning path recommendations, contributing factors, recommendation process, recommendation technique, and LPs evaluation of ontology-based recommender system. The literature review was conducted using the PRISMA flow diagram and the primary studies were identified using a broad automated search method. Nine papers were extracted, and the results are summarized as follows:

1. Ontology-based learning path recommender systems are emerging, although they're not as popular as learning object recommender systems. In addition to being used in CAI, ITS, and VLE, learning path recommendations could also be used in LMS and AR.
2. Learning path classifications include those based on curriculum sequencing approach, pedagogical approach, and recommendation process. The current trend for LP recommendations process is *semi-dynamic* and *dynamic*.
3. Learning path can be recommended either at the individual (individual sequencing) or group level (social sequencing). The factors that influence LP recommendations are student model (e.g., student profile, learner style, competence level), LO (e.g., learning objectives, relationship among concepts, content format), learning activities (e.g., number of accessed LOs, number of assessment trials), and external factors (e.g., similar student opinion, weather).
4. Four phases compose the interaction between students and learning paths: predelivery of the first LO, current LO delivery, LO postdelivery, and predelivery of the next LO. The main step for LP recommendation lies in the third and fourth phases. The current recommendation technique combines several techniques/methods, such as Bayesian networks, data mining, and artificial intelligence. In the future, ontology could collaborate with education, educational psychology, and evolutionary computation.
5. In the existing studies, learning path performances were tested dynamically (i.e., tested with real users) by using control groups in parallel or serial experiments. Tests usually involved 70–90 participants and spanned four months. Future studies could expand the testing type and scale, for example, by applying static testing, using other learning domains, and utilizing comprehensive tests.

Acknowledgements The authors gratefully acknowledge Publisher and Publication Board of Universitas Gadjah Mada for proofreading funding.

Data availability The data file for this systematic review will be made available upon reasonable academic request by the corresponding author upon acceptance.

Declarations

Conflicts of interest/Competing interests The authors have no conflicts of interest to declare that are relevant to the content of this article.

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Authors and Affiliations

Nur W. Rahayu[1,2] · Ridi Ferdiana[2] · Sri S. Kusumawardani[2]

Nur W. Rahayu
nnur@uii.ac.id

Ridi Ferdiana
ridi@ugm.ac.id

[1] Department of Informatics, Universitas Islam Indonesia, Yogyakarta, Indonesia

[2] Department of Electrical Engineering and Information Technology, Universitas Gadjah Mada, Yogyakarta, Indonesia